

Wald Chapter 6: The Schwarzschild Solution

R. M. Wald, General Relativity, Chapter 6; selected worked problems and related solutions.

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General Relativity

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Problem prompts are paraphrased from Robert M. Wald, General Relativity (University of Chicago Press, 1984); the equations, notation, numbering, and guidance follow the source. See the official publisher page.

1. Chapter 6, Problem 1: Isotropic Coordinates

Let M be a three-dimensional manifold with a spherically symmetric Riemannian metric, and suppose $\nabla_a r \neq 0$, where r is defined in (6.1.3).

- a) Show that one may introduce an isotropic radial coordinate $\tilde{r}$ for which

$$ds^2 = H(\tilde{r}) [d\tilde{r}^2 + \tilde{r}^2 d\Omega^2]. \quad (1.1)$$

Thus every spherically symmetric three-dimensional space is conformally flat.

Solution. Assuming the metric can be written in a conformally flat form, then we must have

$$h(r)dr^2 = H(\tilde{r})d\tilde{r}^2, \quad H(\tilde{r})\tilde{r}^2 = r^2, \quad (1.2)$$

which gives the differential equation

$$\frac{\sqrt{h(r)}}{r} dr = \frac{d\tilde{r}}{\tilde{r}} \implies \tilde{r} = A \exp \left(\int^r dr' \frac{\sqrt{h(r')}}{r'} \right). \quad (1.3)$$

- b) Express the Schwarzschild metric in these coordinates and obtain

$$ds^2 = -\frac{(1 - M/2\tilde{r})^2}{(1 + M/2\tilde{r})^2} dt^2 + \left(1 + \frac{M}{2\tilde{r}}\right)^4 [d\tilde{r}^2 + \tilde{r}^2 d\Omega^2]. \quad (1.4)$$

Solution. For the Schwarzschild metric,

$$\tilde{r} = A \exp \left(\int^r dr' \frac{1}{\sqrt{r'^2 - 2Gmr'}} \right) = 2A \left(r - GM + \sqrt{r(r - GM)} \right). \quad (1.5)$$

A can be fixed by imposing the metric to be asymptotically flat, that is as $r \rightarrow \infty$, $H(\tilde{r}) \rightarrow 1$. Therefore, we must have $A = 1/4$. Hence, playing around with the algebra slightly, we find

$$\tilde{r} = \tilde{r} \left(1 + \frac{GM}{2\tilde{r}} \right)^2, \quad (1.6)$$

which gives

$$ds^2 = -\frac{(1 - M/2\tilde{r})^2}{(1 + M/2\tilde{r})^2} dt^2 + \left(1 + \frac{M}{2\tilde{r}}\right)^4 [d\tilde{r}^2 + \tilde{r}^2 d\Omega^2]. \quad (1.7)$$

2. Chapter 6, Problem 3: The Reissner–Nordström Solution

Consider the source-free Maxwell equations (4.3.12) and (4.3.13), with $j^a = 0$, on the static, spherically symmetric spacetime (6.1.5).

- a) Argue that a Maxwell tensor with the same static and spherical symmetries must have the form

$$F_{ab} = 2A(r)(e_0)_{[a}(e_1)_{b]} + 2B(r)(e_2)_{[a}(e_3)_{b]}, \quad (2.1)$$

where the $(e_\mu)_a$ are defined in (6.1.6).

Solution. Since F is a two form, it should in general take the form

$$F = F_{ab}\hat{\theta}^a \wedge \hat{\theta}^b, \quad (2.2)$$

where $\hat{\theta}^a$ is the dual basis of $\hat{e}_a$. Now, excluding the basis that are not invariant under $SO(3)$ transformation, we are left with

$$F = A(r)\hat{\theta}^0 \wedge \hat{\theta}^1 + B(r)\hat{\theta}^2 \wedge \hat{\theta}^3, \quad (2.3)$$

where A, B are only functions of r by the requirement of the universe being static.

- b) Restrict to $B(r) = 0$ and solve Maxwell's equations to find

$$A(r) = -\frac{q}{r^2}, \quad (2.4)$$

interpreting q as the total charge. The branch with $B(r) \neq 0$ is a duality rotation of this electric solution and represents a magnetic monopole.

Solution. Writing the field strength two form in terms of the coordinate basis, we have

$$F = A(r)f^{1/2}h^{1/2}dt \wedge dr. \quad (2.5)$$

Since A, f, h are all functions of r , $dF = 0$ gives no extra informations. The wedge of F is given by

$$\star F = A(r)r^2 \sin \theta d\theta \wedge d\phi, \quad (2.6)$$

up to some constant that would not affect our calculation here. Then, the source-free Maxwell equation gives

$$\partial_r(A(r)r^2) = 0 \implies A(r) = -\frac{q}{r^2}. \quad (2.7)$$

- c) Insert the electromagnetic stress-energy tensor from part (b) into

$$G_{ab} = 8\pi T_{ab}, \quad (2.8)$$

solve the resulting equations, and show that the general metric is

$$ds^2 = -\left(1 - \frac{2M}{r} + \frac{q^2}{r^2}\right) dt^2 + \left(1 - \frac{2M}{r} + \frac{q^2}{r^2}\right)^{-1} dr^2 + r^2 d\Omega^2. \quad (2.9)$$

Solution. The resulting energy momentum tensor in vierbein basis is

$$T_{00} = \frac{1}{8\pi} \frac{q^2}{r^4}, \quad (2.10)$$

$$T_{11} = -\frac{1}{8\pi} \frac{q^2}{r^4}, \quad (2.11)$$

$$T_{22} = \frac{1}{8\pi} \frac{q^2}{r^4}, \quad (2.12)$$

and

$$T_{33} = \frac{1}{8\pi} \frac{q^2}{r^4}. \quad (2.13)$$

The Einstein equation is given by

$$R_{00} + \frac{1}{2}R = 8\pi T_{00}, \quad (2.14)$$

$$R_{11} - \frac{1}{2}R = 8\pi T_{11}, \quad (2.15)$$

$$R_{22} - \frac{1}{2}R = 8\pi T_{22}, \quad (2.16)$$

and

$$R_{33} - \frac{1}{2}R = 8\pi T_{33}, \quad (2.17)$$

where the Ricci tensor is given in (6.1.35-6.1.37). Combining (2.14) and (2.15), we find

$$R_{00} + R_{11} = 0, \quad (2.18)$$

which is equivalent to (6.1.38), and hence

$$f = h^{-1}. \quad (2.19)$$

Note that (2.16) and (2.17) are completely identical, plugging $f = 1/h$ in one of them, we find

$$f'' + \frac{2f'}{r} = \frac{2q^2}{r^4}, \quad (2.20)$$

where we have used

$$R = -R_{00} + R_{11} + 2R_{22}. \quad (2.21)$$

This gives

$$f = h^{-1} = 1 - \frac{C}{r} - \frac{q^2}{r^2}. \quad (2.22)$$

3. Chapter 6, Problem 4: Local Force and Redshift

Let (M, g_{ab}) be stationary with timelike Killing field ξ^a , and define

$$V^2 = -\xi^a \xi_a. \quad (3.1)$$

a) For a stationary observer, show that the acceleration $a^b = u^a \nabla_a u^b$ is

$$a^b = \nabla^b \ln V. \quad (3.2)$$

Solution.

$$a^b = u^a \nabla_a u^b = \frac{u^a \nabla_a \xi^b}{V} - \frac{u^a \xi^b \nabla_a V}{V^2}. \quad (3.3)$$

The second term vanishes by

$$\frac{u^a \xi^b \nabla_a V}{V^2} = -\frac{2\xi^a \xi^b \xi^c \nabla_a \xi_c}{V^3} = 0 \quad (3.4)$$

by $\nabla_a \xi_b = -\nabla_b \xi_a$. Then, we have

$$a^b = \frac{\xi^a \nabla_a \xi^b}{V^2} = -\frac{\xi^a \nabla^b \xi_a}{V^2} = \frac{1}{2} \nabla^b \log V^2 = \nabla^b \log V. \quad (3.5)$$

- b) Suppose additionally that the spacetime is asymptotically flat. Thus there are coordinates t, x, y, z , with $\xi^a = (\partial/\partial t)^a$, in which g_{ab} tends to $\text{diag}(-1, 1, 1, 1)$ as $r = (x^2 + y^2 + z^2)^{1/2} \rightarrow \infty$. See Chapter 11 for further discussion of asymptotic flatness.

A particle of mass m has energy at infinity

$$E = -m\xi^a u_a. \quad (3.6)$$

Hold the particle stationary with a massless string whose far end is held by a stationary observer at large r . If F is the force applied locally to the particle, part (a) gives

$$F = mV^{-1} (\nabla_a V \nabla^a V)^{1/2}. \quad (3.7)$$

Use conservation of energy to show that the force supplied at infinity is

$$F_\infty = VF. \quad (3.8)$$

Interpret the factor V as the redshift relating the two force magnitudes.

Solution. The local magnitude of force is given by

$$F = m\sqrt{a^a a_a} = mV^{-1} (\nabla_a V \nabla^a V)^{1/2}, \quad (3.9)$$

while the energy of a stationary particle measured by a local stationary observer is given by

$$E = m. \quad (3.10)$$

Assuming the stationary particle is pushed through a proper distance δs adiabatically with a constant force F . Then, the work done locally is equal to the local change of energy is given by

$$F\delta s = \delta E. \quad (3.11)$$

For an observer at spatial infinity, he agrees on the proper distance, while the force seen by this observer, F_∞ , is different. Therefore, the change of energy seen by this observer is

$$\delta E_\infty = F_\infty \delta s. \quad (3.12)$$

By the energy redshift, we have

$$V\delta E = \delta E_\infty. \quad (3.13)$$

Therefore, we have

$$F_\infty = VF. \quad (3.14)$$

4. Chapter 6, Problem 5: Relativistic Time Delay

Derive the general-relativistic time-delay formula (6.3.45). *Draft solution: the derivation below does not yet display the final time-delay formula (6.3.45).*

Solution. By time translational symmetry, the time delay from the Earth to the planet in coordinate time must be equivalent to

$$\Delta t = \int_{R_0}^{R_E} dr \frac{1}{(1 - 2GM/r)\sqrt{1 - (1 - 2GM/r)b^2/r^2}} + \int_{R_0}^{R_p} dr \frac{1}{(1 - 2GM/r)\sqrt{1 - (1 - 2GM/r)b^2/r^2}}. \quad (4.1)$$

To first order in G , this is

$$\Delta t = \Delta t \Big|_{G=0} + \frac{\partial(\Delta t)}{\partial G} \Big|_{G=0} G + \mathcal{O}(G^2). \quad (4.2)$$

The first term is simply the time elapsed when the star is absent. Therefore, it must be

$$\Delta t \Big|_{G=0} = (R_E^2 - R_0^2)^{1/2} + (R_p^2 - R_0^2)^{1/2} \quad (4.3)$$

by simple geometry. Recall that

$$R_0^3 - b^2(R_0 - 2GM) = 0 \implies b = R_0 + GM + \mathcal{O}(G^2). \quad (4.4)$$

Then, for the second term, we find

$$\begin{aligned} \frac{\partial(\Delta t)}{\partial G} \Big|_{G=0} = & \frac{2GM}{R_0} \log \left(\frac{r}{R_0} + \frac{\sqrt{r^2 - R_0^2}}{R_0} \right) \Big|_{R_0}^{R_E} + \frac{2GM}{R_0} \log \left(\frac{r}{R_0} + \frac{\sqrt{r^2 - R_0^2}}{R_0} \right) \Big|_{R_0}^{R_p} \\ & - \frac{2GM r}{\sqrt{r^2 - b^2}} \Big|_{R_0}^{R_E} - \frac{2GM r}{\sqrt{r^2 - b^2}} \Big|_{R_0}^{R_p}. \end{aligned} \quad (4.5)$$

Note that special attention is required when substituting R_0 into the last two terms. Carefully keeping terms of first order in G , one would eventually find the residue terms of (6.3.45).

5. Chapter 6, Problem 6: Lifetime Inside the Schwarzschild Horizon

Show that every particle in region II, $r < 2M$, of the extended Schwarzschild spacetime in Figure 6.9, whether geodesic or accelerated, must obey

$$\left| \frac{dr}{d\tau} \right| \geq \left(\frac{2M}{r} - 1 \right)^{1/2}. \quad (5.1)$$

Deduce that no observer in region II can have a remaining proper lifetime greater than

$$\tau = \pi M \sim 10^{-5} \left(\frac{M}{M_\odot} \right) \text{s} \quad (5.2)$$

before reaching $r = 0$. Finally, show that this bound is approached by free fall from $r = 2M$ in the limit $E \rightarrow 0$.

Solution. Directly from the Schwarzschild metric, we immediately

$$\left| \frac{dr}{d\tau} \right| \geq \left| 1 - \frac{2GM}{r} \right|^{1/2} = \left(\frac{2GM}{r} - 1 \right)^{1/2} \quad (5.3)$$

for $r < 2GM$. This gives

$$\tau \leq \int_0^{2GM} \frac{dr}{\sqrt{2GM/r - 1}} = \pi GM. \quad (5.4)$$

The integral is precisely the proper time for a particle to fall to the singularity, released from the event horizon with $E = 0$.